Adult Polymorphic Wide-Complex Tachycardia

AC-08 | Adult Cardiac

Quick flow

01 Confirm Pulse + Polymorphic Wide Rhythm [EMT / AEMT / Paramedic]

Beat-to-beat QRS variation • pads immediately • airway/ventilation • oxygen if hypoxemic • do not delay shock

02 Sustained Polymorphic VT [EMT / AEMT / Paramedic]

Treat as unstable regardless of apparent perfusion

03 Immediate Unsynchronized Shock [Paramedic]

200 J biphasic • verify manufacturer setting • escalate according to device • never attempt synchronization

04 Pulse Lost? [EMT / AEMT / Paramedic]

Begin CPR immediately and open AC-09 VF / Pulseless VT

05 Rhythm Terminates [EMT / AEMT / Paramedic]

Obtain rhythm strip/12-lead • determine whether baseline QT is prolonged

06 Long QT / Torsades Recurrence [Paramedic]

Magnesium sulfate 1-2 g IV over 15 min • may repeat once

07 Normal QT Recurrence [Paramedic]

Treat suspected myocardial ischemia • lidocaine 100 mg IV/IO initial dose

08 Reassess + Transport [EMT / AEMT / Paramedic]

Pads remain attached • continuous monitoring • early notification • anticipate recurrence or VF

Overview

- Adult pathway for patients 16 years of age and older with sustained or recurrent polymorphic wide-complex tachycardia and a pulse.
- All sustained polymorphic ventricular tachycardia is electrically unstable and requires immediate unsynchronized defibrillation.
- Claiborne EMS uses an initial unsynchronized biphasic shock of 200 J; confirm the manufacturer-recommended setting and escalate according to the device.

EMT Care

Rapid recognition, immediate pad placement, supportive care, and preparation for deterioration.

EMT

- Perform the primary assessment, confirm a pulse, support airway and ventilation, and give oxygen only when hypoxemic.
- Recognize polymorphic wide-complex tachycardia by beat-to-beat variation in QRS morphology and immediately request paramedic intervention.
- Apply defibrillation pads, obtain complete vital signs when feasible, and do not delay definitive treatment for a 12-lead ECG.
- Prepare for immediate CPR and AED use if the patient becomes pulseless.
- After rhythm termination, obtain a rhythm strip and 12-lead ECG when trained and equipped without delaying transport.

AEMT Care

All EMT care plus vascular access, medication preparation, and continuous reassessment.

AEMT

- Perform all EMT actions and establish IV/IO access without delaying defibrillation.
- Prepare magnesium sulfate, lidocaine, midazolam, suction, ventilation equipment, and airway equipment for paramedic use.
- Reassess pulse, blood pressure, mental status, perfusion, respiratory status, and rhythm after every shock or clinical change.
- Keep pads attached and prepare for recurrent polymorphic VT, ventricular fibrillation, or cardiac arrest.
- Do not independently administer magnesium sulfate, lidocaine, or procedural sedation under AC-08.

Paramedic Care

All AEMT care plus immediate manual defibrillation, QT-directed medication treatment, and sedation when feasible.

Paramedic

- Perform all EMT and AEMT actions and confirm a pulse with polymorphic wide-complex tachycardia.
- For sustained polymorphic VT, immediately deliver an unsynchronized biphasic shock at 200 J; confirm the manufacturer-recommended setting and escalate according to the device.
- Never attempt synchronized cardioversion for polymorphic VT because reliable synchronization is not possible.
- If the patient is conscious, sedate when immediately feasible with midazolam 2-5 mg IV/IO using the smallest effective dose, but never delay defibrillation.
- After rhythm termination, determine whether the baseline QT is prolonged using the preceding or post-conversion rhythm strip/12-lead ECG.
- For recurrent long-QT polymorphic VT/torsades, give magnesium sulfate 1-2 g IV over 15 minutes; may repeat once for continued recurrence.
- For recurrent polymorphic VT with a normal QT, treat suspected myocardial ischemia and give lidocaine 100 mg IV/IO as the initial dose.

Indications

- Sustained polymorphic wide-complex tachycardia with a pulse in a patient 16 years of age or older.
- Recurrent self-terminating polymorphic VT requiring prevention of recurrence after pads are applied and immediate-shock readiness is established.
- Suspected torsades de pointes associated with a prolonged baseline QT interval.

Contraindications

- There is no contraindication to immediate unsynchronized defibrillation for sustained polymorphic VT.
- Do not use magnesium routinely for polymorphic VT when the baseline QT interval is normal.
- Do not use lidocaine as the torsades/long-QT treatment pathway.

Assessment | Immediate rhythm and perfusion assessment

- Confirm the presence of a pulse and identify beat-to-beat variation in QRS morphology.
- Treat sustained polymorphic VT as electrically and hemodynamically unstable even when a pulse or transient perfusion is present.
- Apply pads immediately and do not delay defibrillation for vascular access, medication administration, sedation, or a 12-lead ECG.
- If no pulse is present or the pulse is lost, begin CPR and immediately open AC-09.

Assessment | QT classification after rhythm termination

- Use the preceding or post-conversion rhythm strip/12-lead ECG to determine whether the baseline QT is prolonged.
- Torsades is suggested by prolonged QT, pause-dependent onset, associated bradycardia, recurrent self-termination, congenital long-QT history, or QT-prolonging medication exposure.
- Normal-QT polymorphic VT is commonly associated with acute myocardial ischemia or infarction.

Assessment | Cause assessment

- Review prescribed and nonprescribed medications for QT prolongation and discontinue suspected QT-prolonging agents.
- Assess for myocardial ischemia, renal failure, dialysis, toxicologic exposure, stimulant use, electrolyte abnormality, congenital long-QT syndrome, and digitalis toxicity.
- Do not delay recurrent shock treatment while investigating the underlying cause.

Treatment steps

- Apply defibrillation pads immediately and treat all sustained polymorphic VT as unstable.
- Deliver an immediate unsynchronized biphasic shock at 200 J; confirm the manufacturer-recommended setting and escalate according to the device.
- Never attempt synchronized cardioversion for polymorphic VT.
- If the rhythm persists or sustained polymorphic VT recurs, repeat unsynchronized defibrillation using the manufacturer-recommended setting and energy escalation.
- If the patient becomes pulseless, begin CPR immediately and open AC-09 VF / Pulseless VT.
- If the patient is conscious, sedate when immediately feasible with midazolam 2-5 mg IV/IO using the smallest effective dose, but never delay defibrillation.
- After rhythm termination, obtain a rhythm strip and 12-lead ECG and determine whether the baseline QT is prolonged.
- For recurrent long-QT polymorphic VT/torsades, give magnesium sulfate 1-2 g IV over 15 minutes; may repeat once for continued recurrence.
- Discontinue suspected QT-prolonging medications.
- For recurrent torsades associated with bradycardia or pauses, contact Medical Control for overdrive pacing.
- For recurrent polymorphic VT with a normal QT, treat suspected myocardial ischemia and give lidocaine 100 mg IV/IO as the initial dose.
- Keep pads attached, reassess complete vital signs, notify early, and transport rapidly while anticipating recurrent polymorphic VT or VF.

Medications

Magnesium sulfate

1-2 g IV over 15 minutes; may repeat once

- Paramedic standing order for recurrent polymorphic VT associated with a prolonged QT interval/torsades de pointes.
- Defibrillation takes priority for any sustained episode.
- Do not administer routinely when the QT interval is normal.
- Monitor continuously for hypotension, bradycardia, respiratory depression, and recurrent dysrhythmia.

Lidocaine

100 mg IV/IO initial dose

- Paramedic standing order for recurrent polymorphic VT only when the baseline QT interval is normal.
- Treat associated myocardial ischemia and keep pads attached.
- Do not substitute lidocaine for immediate defibrillation of a sustained episode.

Midazolam

2-5 mg IV/IO; use the smallest effective dose

- Paramedic sedation when a conscious patient requires unsynchronized defibrillation.
- Do not delay shock delivery for sedation.
- Continuous airway, respiratory, SpO_2, cardiac, and blood-pressure monitoring is required.

Warnings / precautions

- All sustained polymorphic VT is unstable and requires immediate unsynchronized defibrillation.
- Do not attempt synchronized cardioversion; polymorphic QRS complexes cannot be synchronized reliably.
- Do not delay defibrillation for vascular access, medication, sedation, rhythm printing, or a 12-lead ECG.
- Do not give adenosine, diltiazem, verapamil, metoprolol, or another AV-nodal blocker for polymorphic wide-complex tachycardia.
- Avoid amiodarone and other QT-prolonging antiarrhythmics when torsades or a prolonged QT is suspected.
- Routine magnesium is not indicated when the QT interval is normal.
- Recurrent polymorphic VT may rapidly degenerate into ventricular fibrillation.

Clinical pearls

- Polymorphic VT varies in QRS shape and axis from beat to beat and may be difficult to distinguish from coarse VF.
- Torsades de pointes is polymorphic VT associated with a prolonged baseline QT; polymorphic VT with a normal QT is commonly ischemic.
- The QT interval should be assessed on the rhythm before or after the episode rather than during polymorphic VT.
- Defibrillation terminates the episode but does not prevent recurrence; subsequent therapy is directed by the baseline QT.
- Keep pads attached throughout transport because recurrence is common.

Special populations | Long-QT polymorphic VT / torsades

- Give magnesium for recurrence after immediate-shock readiness is established.
- Discontinue suspected QT-prolonging medications.
- For recurrent pause-dependent torsades or associated bradycardia, contact Medical Control for overdrive pacing.

Special populations | Normal-QT polymorphic VT

- Suspect acute myocardial ischemia or infarction and open AC-04.
- Use lidocaine rather than the torsades magnesium pathway.
- Seek expert consultation for suspected catecholaminergic polymorphic VT, digitalis toxicity, or another unusual cause.

Related pathways / contacts

AC-04 ACS / STEMI - Open for suspected myocardial ischemia or infarction.

AC-07 Monomorphic Wide-Complex Tachycardia - Open if the rhythm is regular with uniform QRS morphology.

AC-09 VF / Pulseless VT - Open immediately if the pulse is absent or lost.

AM-03 Dialysis / Renal Failure - Open for suspected dialysis-related or renal emergency.

TE-07 Overdose / Toxic Ingestion - Open for suspected medication, stimulant, digitalis, or other toxicologic cause.

Source and references

American Heart Association. 2025 Guidelines for CPR and ECC: Adult Advanced Life Support - Polymorphic Ventricular Tachycardia. | American Heart Association. 2025 Electrical Cardioversion Algorithm. | Tennessee Emergency Medical Services Board. Tennessee EMS Protocol Guidelines, September 2025. | Claiborne EMS medication formulary and related cardiac, renal, and toxicology protocols.

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